NMC 2014 PRIMARY LEVEL PAPER 2 QUESTIONS

Qn. 1. Which of these numbers is closest to 0.00009? 0.00008, 0.000091, 0.00010, 0.0000911, 0.000999

Qn. 2. Given two sequences: $5, 10, 15, 20, 25, \ldots$ and $7, 14, 21, 28, 35, \ldots$, the first number after 30 common to both sequences is 35. The first number after 300, common to both sequences is _____

Qn. 3. Arrange the following 5 morning times in order starting with the earliest: three minutes to nine, quarter past eight, half past nine, twelve minutes to noon, three minutes past eight.

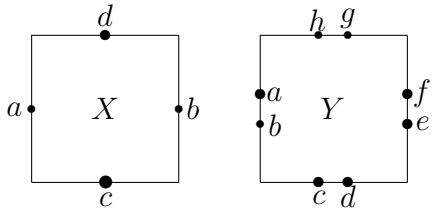
Qn. 4. There are 12 boys in a class. If a quarter of the children in the class are girls, how many children are in the class?

Qn. 5. Five cards are numbered $1, 2, 3, 4, 5$. Two cards are distributed to each of the twins. The sum of the numbers on the two cards the twins are given is 6 for Alice and 7 for Annet. What is the number on the card that was not distributed?

Qn. 6. Four towns are labelled A, B, C, D. The distance between towns A and B is one third the distance between towns A and D. The distance between towns C and D is 60km more than the distance between towns A and B. If the distance between towns B and C is 40km, then the distance between towns A and D is _____

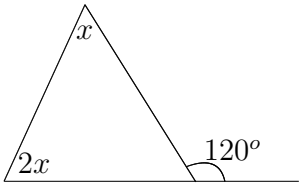
Qn. 7. The average weight of a class of 40 children is 40kg. The average weight of all the 15 girls in the class is 35kg. The average weight of all the boys in the class is _____

Qn. 8. The pair of dots (a, b) divides the square X into two equal parts. The pair of dots (c, d) also divides the square X into two equal parts. Write down all the pairs of dots that divide square Y into two equal parts.



Qn. 9. A primary school had provision of food for 2000 pupils for 90 days. After 30 days, 500 pupils left the school completely due to school fees problems. How many days did the remaining food last?

Qn. 10. In the figure below, angle $x =$ _____ degrees.



SECTION B

Qn. 11. In Sir Apollo Supermarket, the following is the price list for some items:

Item	Price (Shs)
Sugar	233, 500 for 100kgs
Salt	10, 000 for 50kgs
Beans	320, 000 for 100kgs

(a) Peter was given a Shs. 20, 000 note to go and buy 2kgs of Sugar, half a kilo of Salt and 2kgs of Beans. He bought the items. How much change did he take back home.

(b) A retailer buys 300kgs of sugar and sells all at a certain price per kilo. She realises that she has made a profit of Shs. 49, 500 after selling all the 300kgs. At what price did she sell each kilo of sugar?

Qn. 12. In some well organised city, public transport is by bus. The buses start by 6.00am and stop at 10.00pm daily. Every 13minutes, there is a bus at the bus stop which is 2km from Mrs. Daves home. The maximum time the bus waits at the bus stop is 1 minute. Mrs. Daves plans her walk from home in such a way that she does not wait at the bus stop for more than 2minutes before the next bus comes.

(a) If she leaves her home at 7.15am and walks at 8km per hour, which bus will she catch?

(b) If she walks at the same speed, at what time must she leave home to catch the 7.18am bus?

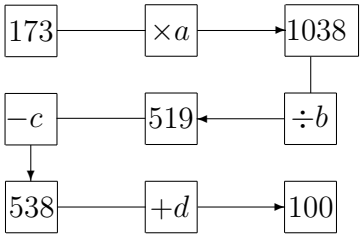
Qn. 13. (a) What is the product of the next two numbers in this sequence:
$12, 45, 15, 42, 18, 39, 21, 36, 24, 33, 27, \ldots, \ldots$?

(b) What is the next term in this series: $G45, H43, I41, J39, K37, L35, \ldots$

Qn. 14. (a) Joan's father was 48 years of age when she was born while her mother was 46 years old when her brother, four years younger to her was born. What is the difference between the ages of her parents?

(b) To make chicken feed, small fish, maize floor, snail shells are mixed in the ration $3 : 5 : 2$ respectively. If Anna uses 50kgs of maize floor, how many kgs of small fish will she use?

Qn. 15. Study the figure below where you start with 173 and end up with 100 after operations of $+, \times, \div$ and $-$. Each of a, b, c, d represents a number required to complete the operation. Find the numbers a, b, c, d.



NMC 2014 PRIMARY LEVEL PAPER 2 SOLUTIONS

SECTION A

Qn1. The distance from 0.00009 is $0.00001, 0.000001, 0.0001, 0.0000011$ and 0.000909. Of these, the smallest distance is 0.000001, therefore 0.000091 is the number closest to 0.00009.

Qn2. In the first sequence, numbers are multiples of 5 so 300 is a member of the first sequence. In the second sequence, numbers are multiples of 7 and so 300 is a member of the this sequence. Therefore 300 is a number common to both sequences. The first number which is a multiple of 5 and 7 greater than 300 is 315.

Qn3. The times are as follows. $8.57am, 8.15am, 9.30am, 11.48am, 8.03am$. Arranging these morning times starting with the earliest, we have $8.03am, 8.15am, 8.57am, 9.30am, 11.48am$.

Qn4. Let the number of children in the class be x. Then

$$\begin{aligned}\tfrac{1}{4}x &= x - 12.\\ \tfrac{1}{4}x - x &= -12.\\ x &= \tfrac{-12\times 4}{-3} = 16.\end{aligned}$$

Therefore there are 16 children in the class.

Qn5. The total weight of a class of 40 children is $40 \times 40 = 1600$ kg. The total weight of all the 15 girls in the class is $15 \times 35 = 525$ kg. The total weight of all the 25 boys in the class is $1600 - 525 = 1075$ kg. Thus the average weight of all the boys in the class is $\frac{1075}{25} = 43$ kg.

Qn6. Possible cards distributed to Alice are $(1, 5)$ and $(2, 4)$. Possible cards distributed to Annet are $(2, 5)$ and $(3, 4)$. The only possibility is for Alice to have $(1, 5)$ and Annet to have $(3, 4)$. In this case the card numbered 2 is not distributed.

Qn7. Let the distance between A and D be x. Then $\frac{1}{3}x + 40 + \frac{1}{3}x + 60 = x$

$$\begin{aligned}\tfrac{2}{3}x - x &= -100\\ \tfrac{-1}{3}x &= -100\\ x &= 300.\end{aligned}$$

Therefore the distance between A and D is 300km.

Qn8. $(a, e), (h, d), (b, f)$ and (c, g).

Note that (a, e) is the same as (e, a) and the same applies to the remaining points.

Qn9. After 30 days, they will have eaten $\frac{30}{90}$ of the food when they are 2000 pupils. The remaining $\frac{60}{90}$ of the food will be eaten by the remaining 1500 pupils. This will take $\frac{2000\times 60}{1500} = 80$ days.

Qn10.

$$x + 2x = 120^\circ$$
$$3x = 120^\circ$$
$$x = 40^\circ.$$

SECTION B

Qn11. (a) Peter spent as follows.

Sugar: $\frac{233500}{100} \times 2 = 4670$.
Salt: $\frac{10000}{50} \times \frac{1}{2} = 100$.
Beans: $\frac{320000}{100} \times 2 = 6400$.

Total money spent $= shs.4670 + shs.100 + shs.6400 = shs.11170$
Therefore Peter took home change of $shs.20000 - shs.11170 = shs.8830$.

(b) Total cost of buying $300kg$ is $233500 \times 3 = shs.700500$. Since the retailer got $shs.49500$ as the profit, the total sales are $shs.700500 + shs.49500 = shs.750000$.
Therefore, the cost per kilo is $\frac{750000}{300} = shs.2500$.

Qn12. (a) Time taken by Wack from home to reach the bus stop is $\frac{\text{Distance}}{\text{speed}} = \frac{2}{8} = \frac{1}{4}$hrs $= 15$ minutes. Therefore, Wack arrived at the bus stop at $7.15 + 0.15 = 7.30$am. At the bus stop, buses arrive at the following times. 6.00am, 6.13am, 6.26am, 6.39am, 6.52am, 7.05am, 7.18am and 7.31am. Hence Wack travelled with the bus that arrived at the bus stop at 7.31am.

(b) $7.18 - 0.15 = 7.03$am. Since she does not want to wait for more than 2minutes, she has to leave home between 7.01am and 7.03am.

Qn13. (a) The next two numbers in the sequence are 30 and 30. Their product is $30 \times 30 = 900$.

(b) The next term in the series is $M33$.

Qn14. (a) Suppose Joan is y years old, then her father is $48 + y$ years old. Her brother is now $y - 4$ years old and so her mother is now $46 + y - 4$ years old. Therefore the difference between the ages of their parents is $(48 + y) - (46 + y - 4) = 6$ years.

(b) Total ratio is $3 + 5 + 2 = 10$. Let x kg be the sack of mixed feed. Then $\frac{5x}{10} = 50$; hence $x = \frac{50 \times 10}{5} = 100$. Thus the amount of small fish used is $\frac{3}{10} \times 100 = 30$kg.

Qn15.

$$173 \times a = 1038 \Longrightarrow a = \frac{1038}{173} = 6.$$

$$1038 \div b = 519 \Longrightarrow b = \frac{1038}{519} = 2.$$

$$519 - -c = 538 \Longrightarrow c = 519 - 538 = -19.$$
$$538 + d = 100 \Longrightarrow d = 100 - 538 = -438.$$